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Electronic device and method of forming electronic device

Abstract

An electronic device includes a first electronic unit, a second electronic unit, a bonding pad, an insulating layer and a circuit layer. The bonding pad is disposed between the first electronic unit and the second electronic unit. The insulating layer is disposed corresponding to the first electronic unit, to the second electronic unit and to the bonding pad. The first electronic unit is electrically connected with the second electronic unit through the circuit layer and through the bonding pad.

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Background/Summary

BACKGROUND OF THE DISCLOSURE
1. Field of the Disclosure
(1) The present disclosure relates to an electronic device and a method of forming an electronic device, in particular to an electronic device and a method of forming the electronic device which may improve its electrical reliability.
2. Description of the Prior Art
(2) In the manufacturing process of an electronic device, it is often necessary to transfer, for example, an electronic unit, such as a known good die (KGD), on a carrier board before an electrical connection step between several electronic units. However, there may be an electrical reliability issue between several electronic units. For example, after several chips are transferred, the position may shift off a predetermined region due to subsequent processes, which may cause electrical problems.
(3) In view of these, it is an urgent issue to provide a method for forming an electronic device which may reduce the electrical reliability problem which is caused by position shift.
SUMMARY OF THE DISCLOSURE
(4) Some embodiments of the present disclosure provide an electronic device. The electronic device may include a first electronic unit, a second electronic unit, a bonding pad, an insulating

layer, and a circuit layer. The bonding pad is disposed between the first electronic unit and the second electronic unit. The insulating layer is disposed corresponding to the first electronic unit, to the second electronic unit and to the bonding pad. The first electronic unit is electrically connected to the second electronic unit through the circuit layer and through the bonding pad.
(5) Some embodiments of the present disclosure further provide a method of forming an electronic device. First, a substrate is provided. The substrate includes a bonding pad, a first electronic unit and a second electronic unit. Then, an insulating layer is provided on the substrate. The insulating layer surrounds the bonding pad, the first electronic unit and the second electronic unit. The bonding pad is disposed between the first electronic unit and the second electronic unit, and the bonding pad is electrically connected to the first electronic unit and to the second electronic unit.
(6) These and other objectives of the present disclosure will no doubt become obvious to those of ordinary skill in the art after reading the following detailed description of the embodiment that is illustrated in the various figures and drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1**, FIG. **2**, FIG. **3**, FIG. **4**, FIG. **5** and FIG. **6** are schematic flowcharts of some embodiments of a method for forming an electronic device according to the present disclosure, which are respectively schematic top views of an electronic device.
(2) FIG. **1**A is a schematic top view of a variant example of another embodiment of a method of forming an electronic device according to the present disclosure.
(3) FIG. **5**A is a schematic cross-sectional view of the electronic device according to some embodiments of the present disclosure along the line A-A′ in FIG. **5**.
(4) FIG. **6**, FIG. **6**A and FIG. **6**B respectively illustrate a schematic partial top view of different embodiments of the bonding pad in accordance with the present disclosure.
DETAILED DESCRIPTION
(5) The present disclosure may be understood by reference to the following detailed description, taken in conjunction with the drawings as described below. It is noted that, for purposes of illustrative clarity and being easily understood by the readers, various drawings of this disclosure show a portion of the electronic device, and certain elements in various drawings may not be drawn to scale. In addition, the number and dimension of each device shown in drawings are only illustrative and are not intended to limit the scope of the present disclosure.
(6) Certain terms are used throughout the description and following claims to refer to particular components. As one skilled in the art will understand, electronic equipment manufacturers may refer to a component by different names. This document does not intend to distinguish between components that differ in name but not function.
(7) In the following description and in the claims, the terms “include”, “comprise” and “have” are used in an open-ended fashion, and thus should be interpreted to mean “contain, but not limited to”.
(8) When a component or a film layer is referred to as “disposed on another component or another film layer” or “connected to another component or another film layer”, it can mean that the component or film layer is directly disposed on another component or film layer, or directly connected to another component or film layer, or there may be other components or film layers in between. In contrast, when a component is said to be “directly disposed on another component or film” or “directly connected to another component or film”, there is no component or film between the two. When a component or a film layer is referred to as “coupled to” another component or another film layer, it can mean that the component or film layer is directly connected to another component or film layer, or indirectly connected to another component or film layer via one or more components. And the terms of joining and connecting may also include a case where both

structures are movable, or both structures are fixed. Furthermore, the terms “electrically connected” or “electrically coupled” may include any direct and indirect means of electrical connection.
(9) Although terms such as first, second, third, etc., may be used to describe diverse constituent elements, such constituent elements are not limited by the terms. The terms are used only to discriminate a constituent element from other constituent elements in the specification. The claims may not use the same terms, but instead may use the terms first, second, third, etc. with respect to the order in which an element is claimed. Accordingly, in the following description, a first constituent element may be a second constituent element in a claim.
(10) The technical features in different embodiments described in the following may be replaced, recombined, or mixed with one another to constitute another embodiment without departing from the spirit of the present disclosure.
(11) FIG. **1**, FIG. **2**, FIG. **3**, FIG. **4**, FIG. **5** and FIG. **6** are schematic flowcharts of some embodiments of a method for forming an electronic device according to the present disclosure, which are respectively schematic top views of an electronic device **100**. In the present disclosure, the electronic device **100** may include an antenna device, a display device, a radar device, a lidar device, a packaging element, a packaging module, etc., but the present disclosure is not limited thereto. The packaging element may include, for example, a system-in-package (SiP) or a system-on-chip (SoC) and other structures, but the present disclosure is not limited thereto. If the electronic device **100** is used in packaging, it may be applied to packaging, for example a wafer level packaging (WLP) or a panel level packaging (PLP), such as a chip-first or a redistribution-layer-first (RDL-first) packaging method, but the present disclosure is not limited thereto. It should be noted that, the electronic device may be any combination of the aforementioned device, but the present disclosure is not limited thereto. Each embodiment of the present disclosure may illustrate a combination of one or more electronic units, bonding pads, insulating layers, and circuit layers on a temporary carrier, wherein the electronic unit of the present disclosure may be a fan-out packaged semiconductor unit as an example, but the present disclosure is not limited thereto. Hereinafter, a display device may be used as an example of the electronic device of the present disclosure, but the present disclosure is not limited thereto.
(12) As shown in FIG. **1** to FIG. **6**, a substrate **101** is provided. The substrate **101** may be a temporary supporting carrier for temporarily supporting one or more electronic units, bonding pads, insulating layers, circuit layers and a combination thereof. For example, in some embodiments, the substrate **101** may include a chip, a wafer, stainless steel, an alloy, carbon fibers, glass fibers, or glass of organic materials, of inorganic materials, or of a combination thereof, but the present disclosure is not limited thereto.
(13) As shown in FIG. **1** to FIG. **6**, the substrate **101** may at least include a first electronic unit **110**, a second electronic unit **120** and a bonding pad **130**, but the present disclosure is not limited thereto. At this time, due to the presence of the substrate **101**, the first electronic unit **110**, the second electronic unit **120** and the bonding pad **130** may be disposed on the upper surface **101**S of the substrate **101** in a substantially coplanar manner. The first electronic unit **110** or the second electronic unit **120** may independently include a passive element or an active element, such as a capacitor, a resistor, an inductor, a sensor, an diode, a transistor, semiconductor element, an integrated circuit (IC), a printed circuit board (PCB) etc. The diode may include a light emitting diode or a photodiode. The light emitting diode may, for example, include an organic light emitting diode (OLED), a mini LED, a micro LED, or a quantum dot LED, but the present disclosure is not limited thereto. The order of providing the first electronic unit **110**, the second electronic unit **120** and the bonding pad **130** on the substrate **101** may be optionally arranged according to the situations as follows.
(14) FIG. **1** illustrates a schematic top view of an embodiment of a method of forming an electronic device **100** according to the present disclosure. For example, as shown in FIG. **1**, in some embodiments, at least one bonding pad **130** may be provided on the upper surface **101**S of the

substrate **101** in advance. For the convenience of description, the following disclosure is an example of providing a bonding pad **130** on the upper surface **101**S of the substrate **101** in advance. For example, in a top view, a bonding pad **130** may be provided at a position of the substrate **101** in advance, and spaces for a first electronic unit (not shown) and for a second electronic unit (not shown) are reserved on the left side and the right side of the bonding pad **130**, but the present disclosure is not limited thereto. Then, as shown in FIG. **2**, in a top view, the first electronic unit **110** and the second electronic unit **120** may be further provided on the upper surface **101**S of the substrate **101** in the presence of the bonding pad **130** so that the bonding pad **130** is disposed between the first electronic unit **110** and the second electronic unit **120** on the upper surface **101**S of the substrate **101**. The manufacturing method of providing the bonding pad **130** on the substrate **101** in advance is beneficial, for example, to facilitate the position limit reference of the first electronic unit (not shown) and the second electronic unit (not shown). The bonding pad **130** is provided on the substrate **101** before the first electronic unit and the second electronic unit are provided so that the bonding pad **130** may be, for example, served as an alignment mark, but the present disclosure is not limited thereto.

(15) FIG. **1**A is a schematic top view of a variant embodiment of a method of forming an electronic device **100** according to the present disclosure. Alternatively, as shown in FIG. **1**A, in some embodiments, the first electronic unit **110** and the second electronic unit **120** may be provided on the substrate **101** in advance. For example, in a top view, the first electronic unit **110** and the second electronic unit **120** may be provided on the upper surface **101**S of the left side and right side of the substrate **101** in advance to reserve space for a bonding pad in the middle of the substrate **101**, but the present disclosure is not limited thereto. Then, as shown in FIG. **2**, a bonding pad **130** may be provided on the upper surface **101**S of the substrate **101** in the presence of the first electronic unit **110** and of the second electronic unit **120** so that the bonding pad **130** is disposed between the first electronic unit **110** and the second electronic unit **120**. The fabrication method of providing the first electronic unit **110** and the second electronic unit **120** on the substrate **101** in advance is advantageous, for example, to facilitate the flexible arrangement of the process. For example, in the process of transferring an electronic unit to the substrate, the transfer may be less susceptible to interference from other components, but the present disclosure is not limited thereto.

(16) When the first electronic unit **110**, the second electronic unit **120** and the bonding pad **130** are respectively provided on the substrate **101**, between the first electronic unit **110** and the bonding pad **130**, or between the second electronic unit **120** and the bonding pad **130** there may be not parallel arrangement. As shown in FIG. **2**, the bonding pad **130** may have an extending direction L1 along the long side **130**L of the bonding pad **130**. Similarly, the side **110**L of the first electronic unit **110** which is close to the long side **130**L of the bonding pad **130** may have an extension direction L2. According to some embodiments of the present disclosure, between the extension direction L1 and the extension direction L2 there may be an angle θ. If the first electronic unit **110** and the bonding pad **130** are not arranged parallel to each other, the angle θ may not be equal to 0. In some embodiments, the angle θ may be greater than or equal to 0. In some embodiments, the angle θ may be smaller than 45°, for example, $0\leq\theta<45°$, but the present disclosure is not limited thereto. According to some embodiments, the angle θ may be greater than or equal to 0 and less than 15°, but the present disclosure is not limited thereto. For example, the angle θ may be detected by an automatic optical system (Auto-Optical Inspection, AOI). When the angle θ is greater than or equal to 0 and less than 45°, it is beneficial to improve the electrical reliability between electronic units, but the present disclosure is not limited thereto.

(17) In some embodiments, the bonding pad **130** may include a conductor layer **131** (shown in FIG. **5**A) disposed corresponding to the base layer **132**. The conductor layer **131** may include a metal conductive material or a non-metallic conductive material with low electric resistance, such as copper, nickel, gold, silver, tin, a transparent conductive material, other suitable materials, or a combination of the above conductive materials for packaging, but the present disclosure is not

limited thereto. The transparent conductive material may include a transparent conductive material such as indium tin oxide (ITO), indium zinc oxide (IZO), and indium gallium zinc oxide (IGZO), but the present disclosure is not limited thereto. In some embodiments, a suitable process may be used to form the conductor layer **131**. A suitable process may include, for example, a plating process, a coating process, other suitable processes, or a combination thereof, but the present disclosure is not limited thereto. The bonding pad **130** of the present disclosure may be used as an auxiliary conductive line in a region, so that two electronic units may be electrically connected through the bonding pad **130** to serve as a connecting component. The bonding pad **130** may include, for example, a quadrangle, a circle or other shapes, but the present disclosure is not limited thereto. In a top view, the bonding pad **130** may include, for example, a rounded corner, an arc-shaped corner, a grid shape or other different implementations. By providing the bonding pad **130** with a rounded corner or an arc-shaped corner, for example, the tip discharge effect may be reduced, and the reliability of the electronic device may be improved, but the present disclosure is not limited thereto.
(18) Next, as shown in FIG. **1** to FIG. **6**, an insulating layer **140** may be further provided on the upper surface **101**S of the substrate **101**. For example, in some embodiments, the insulating layer **140** may be provided so that the insulating layer **140** surrounds the first electronic unit **110**, the second electronic unit **120**, and the bonding pad **130**. In other words, the insulating layer **140** may be disposed corresponding to the first electronic unit **110**, to the second electronic unit **120** and to the bonding pad **130**. According to some embodiments of the present disclosure, the above-mentioned “disposed corresponding to” may refer to the insulating layer **140** in direct contact with at least apart of the corresponding element, for example, in a top view, to an arrangement in direct contact with at least one side of the first electronic unit **110** or with at least one side of the second electronic unit **120**. According to some embodiments, the insulating layer **140** surrounding the first electronic unit **110**, the second electronic unit **120** and the bonding pad **130**, for example, in a cross-sectional view, may refer to the insulating layer **140** in direct contact with at least two sides of the first electronic unit **110**, of the second electronic unit **120** and of the bonding pad **130**. The insulating layer **140** may be a soft, film-like, powder-like or liquid encapsulating material before being provided; for example, may include a pre-polymerized epoxy encapsulating material, but the present disclosure is not limited thereto. After the pre-polymerized epoxy encapsulation material is cured, the cured insulating layer **140** may be formed, but the present disclosure is not limited thereto. For example, the insulating layer **140** in direct contact with the corresponding element is beneficial to prevent moisture or oxygen from contacting or penetrating the corresponding element, that is, the weather resistance of the electronic device may be improved, but the present disclosure is not limited thereto.
(19) Then, as shown in FIG. **1** to FIG. **6**, after the insulating layer **140** is provided on the upper surface **101**S of the substrate **101**, the substrate **101** may be flipped over so that the upper surface **101**S shown in FIG. **2** is disposed upside down. The step of flipping over the substrate **101** may also include the step of removing the temporary substrate **101**. Optionally, the step of flipping over the substrate **101** and the step of removing the substrate **101** may be carried out separately. For example, in some embodiments, the step of flipping over the substrate **101** may be carried out before the step of removing the substrate **101** is carried out. Alternatively, in some embodiments, the step of removing the substrate **101** may be carried out before the step of flipping over the substrate **101** is carried out. After the step of removing the substrate **101**, the bonding terminals **110**C of the first electronic unit **110** and the bonding terminals **120**C of the second electronic unit **120** may be respectively exposed. According to some embodiments of the present disclosure, the bonding pad **130** may be provided to electrically connect the first electronic unit **110** or the second electronic unit **120**. For example, in some embodiments, the bonding pad **130** may be provided to electrically connect the bonding terminals **110**C of the first electronic unit **110** and the bonding terminals **120**C of the second electronic unit **120** so that the first electronic unit **110** may be

electrically connected to the second electronic unit **120** through the bonding pad **130**.
(20) As shown in FIG. **1** to FIG. **6**, in a top view, a circuit layer **150** may be provided on the insulating layer **140** (shown in FIG. **3**), so that the first electronic unit **110** may be electrically connected to the bonding pad **130** through the circuit layer **150** on the insulating layer **140**, and the bonding pad **130** may be electrically connected to the second electronic unit **120** through the circuit layer **150** on the insulating layer **140** for example, so that the bonding terminals **110**C of the first electronic unit **110** may be electrically connected to the bonding terminals **120**C of the second electronic unit **120** through the bonding pad **130**. According to some embodiments of the present disclosure, a suitable process may be used to provide the circuit layer **150** to serve as an electrical wire between the first electronic unit **110** and the second electronic unit **120**. In some embodiments, the circuit layer **150** may be provided by, for example, a photo process combined with a lithography process, an electroplating process, a coating process, other suitable processes, or a combination thereof, but the present disclosure is not limited thereto.
(21) According to some embodiments of the present disclosure, the circuit layer **150** may include a composite layer structure, for example, may include a multi-layer conductive layer and a multi-layer insulating layer, to provide the needed circuit pattern and distribution by the connection between the multi-layer conductive layer and the multi-layer insulating layer. Therefore, the circuit layer **150** may be regarded as a redistribution layer (RDL). The multi-layer conductive layer and the multi-layer insulating layer may stack on one another to form a fan-out panel level packaging (FOPLP), for example, to be beneficial for the electronic device to have higher input/output (I/O) density, or to reduce the size of the electronic device. The conductive layer in the circuit layer **150** may include, for example, a first conductive layer **151**, a second conductive layer **152** and a stud **153**, but the present disclosure is not limited thereto. The materials of the conductive layer in the circuit layer **150** may include, for example, titanium, copper, electroplated copper, other suitable conductive materials, or a combination thereof, but the present disclosure is not limited thereto. On the other hand, the circuit layer **150** may include multiple insulating layers. In FIG. **4** one insulating layer **159** is used to represent one or more insulating layers in the circuit layer **150**. For example, the multi-layer insulating layer may include an organic material or an inorganic material, and the organic material or the inorganic material may include, for example, photosensitive polyimide (PSPI), a build-up film (ABF), silicon oxide (SiOx), silicon nitride (SiNx), Silicon oxynitride (SiOxNy), other suitable insulating materials or a combination thereof, but the present disclosure is not limited thereto. According to some embodiments of the present disclosure, the circuit layer **150** may include an element such as a transistor, a capacitor, a diode, and a resistor, but it is not limited thereto. According to some embodiments of the present disclosure, the first electronic unit **110** may be electrically connected to the conductor layer **131** of the bonding pad **130** through the circuit layer **150** on the insulating layer **140**. According to some embodiments of the present disclosure, the conductive layer **131** of the bonding pad **130** may be electrically connected to the second electronic unit **120** through the second conductive layer **152** of the circuit layer **150**. Electronic units such as the first electronic unit **110** or the second electronic unit **120**, for example, may be disposed between the circuit layer **150** and the insulating layer **140**.
(22) As shown in FIG. **1** to FIG. **6**, in a top view, the circuit layer **150** is provided, so that the first electronic unit **110** may be electrically connected to the bonding pad **130** through the circuit layer **150** on the insulating layer **140**, and after the bonding pad **130** is electrically connected to the second electronic unit **120** through the circuit layer **150** on the insulating layer **140**, the step of dividing the bonding pad **130** may be further carried out, so that the bonding pad **130** may form a plurality of bonding pad regions, for example, divided to form independent bonding pad region **133**, bonding pad region **134** and bonding pad region **135**. There may be a dividing line **136** disposed between adjacent bonding pad regions. According to some embodiments of the present disclosure, suitable processes may be used to divide the bonding pad **130** to form the dividing line **136**, for example, a laser cutting process, an etching process, other suitable processes, or a

combination thereof, but the present disclosure is not limited thereto.

(23) After the above steps, an electronic device **100** according to some embodiments of the present disclosure may be provided. FIG. **5**A is a schematic cross-sectional view of the electronic device **100** according to some embodiments of the present disclosure along the line A-A′ in FIG. **5**, wherein the Z direction is the normal direction of the electronic device **100**. Please refer to FIG. **5** and to FIG. together, the electronic device **100** according to some embodiments of the present disclosure may include a first electronic unit **110**, a second electronic unit **120**, a bonding pad **130**, an insulating layer **140** (in FIG. **5**A), and a circuit layer **150**.

(24) The first electronic unit **110** or the second electronic unit **120** may respectively include a passive element or an active element. Please refer to the above for details. The bonding pad **130** may be disposed between the first electronic unit **110** and the second electronic unit **120**. According to some embodiments of the present disclosure, the bonding pad **130** may include a plurality of bonding pad regions that are not connected to each other or are electrically insulated from each other; for example, may include a plurality of bonding pad region **133**, bonding pad region **134**, and bonding pad region **135** which are independently arranged, but the present disclosure is not limited thereto. There may be a dividing line **136** disposed between adjacent bonding pad regions so that the adjacent bonding pad regions are not connected, but the present disclosure is not limited thereto. In some embodiments, the bonding pad **130** may include at least a conductor layer **131**, and the conductor layer **131** may include a metallic conductive material or a non-metallic conductive material with low electric resistance, such as copper, aluminum, titanium, molybdenum, nickel, gold, silver, tin, a transparent conductive material, other suitable materials, or a combination thereof to serve as a conductive material for packaging, but the present disclosure is not limited thereto. The transparent conductive material may include a transparent conductive material such as indium tin oxide (ITO), indium zinc oxide (IZO), and indium gallium zinc oxide (IGZO), but the present disclosure is not limited thereto. In some embodiments, the bonding pad **130** may further include a base layer **132**, so that the conductor layer **131** may be disposed on the base layer **132**, and the base layer **132** may be disposed between the conductor layer **131** and the insulating layer **140**. The base layer **132** may, for example, facilitate the bottom layer of the bonding pad **130** to be close to the material properties of the insulating layer **140** so that the bonding pad **130** may be easily integrated with the insulating layer **140**. The base layer **132** may include an organic material or an inorganic material, such as silicon, glass, ceramic, plastic, other suitable materials or a combination thereof, but the present disclosure is not limited thereto. In some embodiments, suitable processes may be used to form the conductor layer **131**; for example may include an electroplating process, a coating process, other suitable processes or a combination thereof, but the present disclosure is not limited thereto. In some embodiments, in a schematic cross-sectional view, such as shown in FIG. **5**A, the base layer **132** may have at least one rounded corner C, but the present disclosure is not limited thereto. By the design of the base layer **132** with the rounded corner, the risk of cracking between the interface of the base layer **132** and the insulating layer **140** may be reduced, but the present disclosure is not limited thereto. In some embodiments, the Young's modulus of the base layer **132** may be, for example, between 1000 MPa and 20000 MPa, but the present disclosure is not limited thereto. In some embodiments, the coefficient of thermal expansion (CTE) of the base layer **132** may be, for example, between 3 ppm/° K and 10 ppm/° K, but the present disclosure is not limited thereto. The base layer **132** having the above-mentioned properties may allow the bonding pad **130** including the base layer **132** to have the advantage of being close to the material properties of the insulating layer **140** so that the bonding pad **130** may be easily integrated with the insulating layer **140**. In other words, with the provision of the base layer **132**, the risk of cracking between the bonding pad **130** and the insulating layer **140** may be reduced.

(25) As shown in FIG. **1** to FIG. **6** the bonding pad **130** may include a bonding pad region **133**, a bonding pad region **134**, and a bonding pad region **135**, and the extension line L1 may pass through

one side of the bonding pad region **133**, of the bonding pad region **134** and of the bonding pad region **135**. The side **110**L of the first electronic unit **110** adjacent to the side of the bonding pad region **133**, of the bonding pad region **134** and of the bonding pad area **135** may have an extension line L2. There may be an angle θ between the extension line L1 and the extension line L2. If the first electronic unit **110** and the bonding pad **130** are not arranged parallel to each other, the angle θ may not be equal to 0. According to some embodiments of the present disclosure, the angle θ may be greater than or equal to 0. In some embodiments, when reliability is taken into consideration, the angle θ may be smaller than 45°, for example, 0≤θ<45°, but the present disclosure is not limited thereto. According to some embodiments, the angle θ may be greater than or equal to 0 and less than 15°, but the present disclosure is not limited thereto. For example, the angle θ may be detected by an automatic optical system (Auto-Optical Inspection, AOI). When the angle θ is greater than or equal to 0 and less than 45°, it is beneficial to improve the electrical reliability between electronic units. The sides of the bonding pad region **133**, of the bonding pad region **134**, the side edges of the bonding pad region **135** and the longer side **130**L of the bonding pad **130** may have the same extending direction. With the design of a rounded corner or of an arc-shaped corner of the bonding pad **130**, for example, the tip discharge effect may be reduced, to advantageously improve the reliability of the electronic device, but the present disclosure is not limited thereto.

(26) The bonding pad **130** of the present disclosure may have various different implementations. FIG. **6**, FIG. **6**A and FIG. **6**B respectively illustrate a schematic partial top view of different embodiments of the bonding pad **130** corresponding to FIG. **5** in accordance with the present disclosure, wherein the extension line L1 is parallel to the Y direction, the X direction is perpendicular to the Y direction, and the X direction and the Y direction are respectively perpendicular to the Z direction shown in FIG. **5**A. As shown in FIG. **6**, in some embodiments, the bonding pad **130** of the present disclosure may have a shape of a solid structure; in other words, the bonding pad **130** of the present disclosure may be a complete piece of linear pattern or a complete piece of rectangular conductive material, generally with smooth sides or with rounded corners. It may be optionally patterned to form a plurality of bonding pad regions, but the present disclosure is not limited thereto. Or as shown in FIG. **6**A, in some embodiments, the bonding pad **130** of the present disclosure may have a shape of hollow structures; in other words, the bonding pad **130** of the present disclosure may be provided with grid patterns. For example, the bonding pad **130** may be composed of a plurality of parts, such as a nm part P1 roughly in a rectangular shape or in a square shape, a long strip-shaped part P2 having a first oblique direction, and a long strip-shaped part P3 having a second oblique direction R. The first oblique direction may be generally orthogonal to the second oblique direction R but is not parallel to the extending direction L1 of the bonding pad **130**, so that the parts P2 and the parts P3 may be interspersed, and an opening **137** may be formed between a part P2 and a part P3. The opening **137** may expose the base layer **132** (please see FIG. **5**A) beneath the conductor layer **131**, or expose the insulating layer **140** (please see FIG. **5**A) beneath the base layer **132** (please see FIG. **5**A). Or as shown in FIG. **6**B, in some embodiments, the bonding pad **130** of the present disclosure may be in the form of stripes; in other words, the bonding pad **130** of the present disclosure may be provided with alternately arranged bonding pad region **133**, bonding pad region **134**, bonding pad region **135** and the dividing line **136** so that the adjacent bonding pad regions are separated by the dividing line **136** and arranged along the extending direction L1. The bonding pad **130** may include an electrically breakable wire type. For example, some portion of the bonding pad **130** may be broken by controlled electric current to form a plurality of bonding pad regions segregated by the dividing lines **136**. A single bonding pad region may be used as a single electrical connection island corresponding to a bonding terminal (please see FIG. **5**) of the first electronic unit **110** (please see FIG. **5**) and to a bonding terminal (please see FIG. **5**) of the second electronic unit **120** (please see FIG. **5**).

(27) As shown in FIG. **5**A, the first electronic unit **110** or the second electronic unit **120** may respectively have an appropriate electronic unit thickness. The electronic unit thickness of the first

electronic unit **110** may be a vertical distance which is measured from the top surface **110**T of the first electronic unit **110** to the bottom surface **110**S of the first electronic unit **110** along the normal direction (Z direction) of the electronic device **100**. Similarly, the electronic unit thickness of the second electronic unit **120** may be a vertical distance which is measured from the top surface **120**T of the second electronic unit **120** to the bottom surface **120**S of the second electronic unit **120** along the normal direction of the electronic device **100**. The thickness of the bonding pad **130** may be a vertical distance which is measured from the top surface **131**S of the bonding pad **130** to the bottom surface of the bonding pad **130** along its normal direction. If the electronic unit thickness of the first electronic unit **110** and the electronic unit thickness of the second electronic unit **120** are different, the maximum thickness of the electronic unit thickness of the first electronic unit **110** and the second electronic unit **120** is referred to as the maximum thickness T1 (the first electronic unit **110** shown in the figure has the maximum thickness T1, but the present disclosure is not limited thereto), and the thickness of the bonding pad is T2. According to some embodiments of the present disclosure, the thickness of the electronic unit and the thickness of the bonding pad **130** may satisfy $T1 \geq T2$. The above arrangement may reduce the risk of circuit layer transfer when the thickness difference is too large, but the present disclosure is not limited thereto.

(28) The insulating layer **140** may include a first portion **1401**, a second portion **1402** and a third portion **1403**. The first portion **1401** may correspond to the first electronic unit **110**, the second portion **1402** may correspond to the second electronic unit **120**, and the third portion **1403** may correspond to the bonding pad **130**. The first portion **1401**, the second portion **1402** and the third portion **1403** may respectively have an appropriate height. The height corresponding to the first portion **1401** of the first electronic unit **110** may be a vertical distance which is measured from the top surface **110**T of the first electronic unit **110** to the bottom surface **140**B of the insulating layer **140** along the normal direction (Z direction) of the electronic device **100**. The height corresponding to the second portion **1402** of the second electronic unit **120** may be a vertical distance which is measured from the top surface **120**T of the second electronic unit **120** to the bottom surface **140**B of the insulating layer **140** along the normal direction (Z direction) of the electronic device **100**. The height corresponding to the third portion **1403** of the bonding pad **130** may be a vertical distance which is measured from the top surface **131**S of the bonding pad **130** to the bottom surface **140**B of the insulating layer **140** along the normal direction (Z direction) of the electronic device **100**. The height of the third portion **1403** corresponding to the bonding pad **130** is H1. If the height of the first portion **1401** corresponding to the first electronic unit **110** is different from the height of the second portion **1402** corresponding to the second electronic unit **120**, the maximum height of the first portion **1401** and the second portion **1402** is referred to as the maximum height H2. For example, in some embodiments, the height of the third portion **1403** may be not significantly different from the maximum height of the first portion **1401** and the second portion **1402**. According to some embodiments of the present disclosure, the height H1 of the third portion **1403** and the maximum height H2 of the first portion **1401** and the second portion **1402** may satisfy $0.9 \leq (H1/H2) \leq 1.1$.

(29) A circuit layer **150** may be disposed on the insulating layer **140**, so that the first electronic unit **110** may be electrically connected to the bonding pad **130** through the circuit layer **150** on the insulating layer **140**. In addition, the bonding pad **130** may be electrically connected to the second electronic unit **120** through the circuit layer **150** on the insulating layer **140**. The circuit layer **150** may include a composite layer structure. The circuit layer **150** of the composite layer structure may include multiple conductive layers and multiple insulating layers (represented by the insulating layers **159**), wherein the circuit layer **150** may, for example, at least include a first conductive layer **151**, a second conductive layer **152** and a stud **153**, but the present disclosure is not limited thereto. The stud may be for example UBM (an upper metal bonding pad), and a conductive material may be disposed on the stud. The conductive material may be for example a solder, a Cu pad, an Al pad or other suitable materials, but the present disclosure is not limited thereto. An external electronic

component, such as a PCB board, an IC, a capacitor or other suitable electronic components may be electrically connected to the first electronic unit or to the second electronic unit through the stud, but the present disclosure is not limited thereto. The multi-layer conductive layer and the multi-layer insulating layer may stack on one another to form a fan-out panel level packaging (FOPLP), to be beneficial for the electronic device to have higher input/output (I/O) density, or to reduce the size of the electronic device.

(30) The space between the first conductive layer **151** and the second conductive layer **152** is d. The space is the minimum distance between the first conductive layer **151** and the second conductive layer **152** which are connected to and in contact with the same bonding pad **130**. For example, in a top view, the space d is the minimum distance between an end point of the first conductive layer **151** and an end point of the second conductive layer **152**. The first conductive layer **151** or the second conductive layer **152** may respectively have an appropriate conductive layer width. The conductive layer width of the first conductive layer **151** may be the minimum distance in the vertical direction along the extension direction of the first conductive layer **151** itself. Please refer to the above-mentioned method of the conductive layer width of the first conductive layer **151** for the calculation of the conductive layer width of the second conductive layer **152**. If the conductive layer width of the first conductive layer **151** and the conductive layer width of the second conductive layer **152** are different, the maximum conductive layer width of the first conductive layer **151** and the second conductive layer **152** is referred to as the maximum width W. For example, in some embodiments, to be advantageous to have good electrical insulation, there may be an appropriate space d disposed between the first conductive layer **151** and the second conductive layer **152**. According to some embodiments of the present disclosure, the maximum width W and the space d may satisfy $d \geq (W/2)$, to facilitate the electronic device **100** to have good electrical insulation.

(31) According to some embodiments of the present disclosure, the circuit layer **150** may include a first conductive layer **151** simultaneously in contact with at least a part of a surface of the first electronic unit **110**, for example at least a part of the top surface **110**T of the first electronic unit **110**, at least a part of a surface **140**S of the insulating layer **141**, and at least a part of a surface **131**S of the conductor layer **131** so that the first electronic unit **110** may be electrically connected to the conductor layer **131** of the bonding pad **130** through the circuit layer **150** on the insulating layer **140**. According to some embodiments of the present disclosure, the circuit layer **150** may include a second conductive layer **152** simultaneously in contact with at least a part of a surface of the second electronic unit **120**, for example at least a part of a top surface **120**T of the second electronic unit **120**, at least another part of the surface **140**S of the insulating layer **141**, and at least another part of the surface **131**S of the conductor layer **131** so that the conductor layer **131** of the bonding pad **130** may be electrically connected to the second electronic unit **120** through the second conductive layer **152** of the circuit layer **150**. In other words, the first electronic unit **110** may be bonded to the second electronic unit **120** via the electrical connection to the bonding pad **130**.

(32) According to the method for forming a semiconductor device according to some embodiments of the present disclosure, a connecting element (for example, the bonding pad in the above-mentioned embodiment) may be arranged between two adjacent semiconductor units of an electronic device to reduce the problem of electrical reliability such as abnormal mutual electrical connection caused by the positional deviation of the semiconductor units. Such a connecting element has the advantage of improving the reliability of the electronic device, for example, improving the electrical reliability of the electronic device.

(33) Those skilled in the art will readily observe that numerous modifications and alterations of the device and method may be made while retaining the teachings of the disclosure. Accordingly, the above disclosure should be construed as limited only by the metes and bounds of the appended claims.

Claims

1. An electronic device, comprising: a first electronic unit and a second electronic unit; a bonding pad disposed between the first electronic unit and the second electronic unit, wherein the bonding pad comprises a plurality of bonding pad regions which are not connected to each other; an insulating layer disposed corresponding to the first electronic unit, to the second electronic unit and to the bonding pad; and a circuit layer; wherein, the first electronic unit is electrically connected to the second electronic unit through the circuit layer and through the bonding pad.
2. The electronic device of claim 1, wherein the bonding pad further comprises: a base layer; and a conductor layer disposed corresponding to the base layer.
3. The electronic device of claim 2, wherein the circuit layer comprises a first conductive layer in contact with at least a part of a surface of the first electronic unit, with at least a part of a surface of the insulating layer and with at least a part of a surface of the conductor layer.
4. The electronic device of claim 3, wherein the circuit layer comprises a second conductive layer in contact with at least a part of a surface of the second electronic unit, with at least another part of the surface of the insulating layer and with at least another part of the surface of the conductor layer.
5. The electronic device of claim 2, wherein the base layer comprises at least one rounded corner.
6. The electronic device of claim 1, wherein the circuit layer comprises a first conductive layer and a second conductive layer, a space between the first conductive layer and the second conductive layer is d, a maximum width of one of the first conductive layer and the second conductive layers is W, and $d \geq (W/2)$.
7. The electronic device of claim 1, wherein a maximum thickness of one of the first electronic unit and the second electronic unit is T1, a thickness of the bonding pad is T2, and $T1 \geq T2$.
8. The electronic device of claim 1, wherein the insulating layer comprises a first portion, a second portion and a third portion, the first portion corresponds to the first electronic unit, the second portion corresponds to the second electronic unit, and the third portion corresponds to the bonding pad, a height of the third portion is H1, a maximum height of one of the first portion and the second portion is H2, and $0.9 \leq (H1/H2) \leq 1.1$.
9. The electronic device of claim 1, wherein in a top view of the electronic device, an angle θ is between an extension direction of the bonding pad and an extension direction of an edge of the first electronic unit close to the bonding pad, and $0 \leq \theta < 45°$.
10. A method of forming an electronic device, comprising: providing a substrate comprising a bonding pad, a first electronic unit and a second electronic unit; dividing the bonding pad by laser to form a plurality of bonding pad regions; and providing an insulating layer on the substrate, wherein the insulating layer is correspondingly provided with respect to the bonding pad, to the first electronic unit and to the second electronic unit; wherein, the bonding pad is disposed between the first electronic unit and the second electronic unit, and the bonding pad is electrically connected to the first electronic unit and to the second electronic unit.
11. The method of forming the electronic device of claim 10, wherein the bonding pad is provided on the substrate before the first electronic unit and the second electronic unit are provided on the substrate.
12. The method of forming the electronic device of claim 10, wherein the first electronic unit and the second electronic unit are provided on the substrate before the bonding pad is provided on the substrate.
13. The method of forming the electronic device of claim 10, wherein an extension direction of the bonding pad and an extension direction of a side of the first electronic unit close to the bonding pad have an angle θ, and $0 \leq \theta < 45°$.
14. The method of forming the electronic device of claim 10, further comprising: providing a

circuit layer on the insulating layer, and the first electronic unit is electrically connected to the second electronic unit through the circuit layer and through the bonding pad.

15. The method of forming the electronic device of claim 14, wherein the circuit layer comprises a first conductive layer in contact with at least a part of a surface of the first electronic unit, with at least a part of a surface of the insulating layer and with at least a part of a surface of a conductor layer included in the bonding pad.

16. The method of forming the electronic device of claim 14, wherein the circuit layer comprises a first conductive layer and a second conductive layer, a space between the first conductive layer and the second conductive layer is d, a maximum width of one of the first conductive layer and the second conductive layers is W, and $d \geq (W/2)$.

17. The method of forming the electronic device of claim 10, wherein a maximum thickness of one of the first electronic unit and the second electronic unit is T1, a thickness of the bonding pad is T2, and $T1 \geq T2$.

18. The method of forming the electronic device of claim 10, wherein the insulating layer comprises a first portion, a second portion and a third portion, the first portion corresponds to the first electronic unit, the second portion corresponds to the second electronic unit, and the third portion corresponds to the bonding pad, a height of the third portion is H1, a maximum height of one of the first portion and the second portion is H2, and $0.9 \leq (H1/H2) \leq 1.1$.